

1. Purpose

This document lists every validation in the Presiding Officer (PO) 2-2 hourly turnout flow (urban / rural). Use it for QA recheck after builds.

2. Area type (Urban vs Rural)

- Source: PO login UrbanRural -> U/URBAN = Urban, R/RURAL = Rural.
- Unknown / missing area type is treated as Rural (no 5 PM slot).
- Urban slots: 9 AM, 11 AM, 1 PM, 3 PM, 5 PM, Queue, Final.
- Rural slots: 9 AM, 11 AM, 1 PM, 3 PM, Queue, Final (no 5 PM).

3. Mandatory last slot before next cards

Step 1: Urban - 5 PM mandatory gate

- User must fill and SAVE 5 PM details.
- Until 5 PM is saved: Queue and Final cards stay LOCKED / disabled.
- Tapping a locked card shows snackbar: fill and save 5 PM first.
- Repository also blocks queue/final save if 5 PM not saved.

Step 2: Rural - 3 PM mandatory gate

- User must fill and SAVE 3 PM details.
- Until 3 PM is saved: Queue and Final cards stay LOCKED / disabled.
- Tapping a locked card shows snackbar: fill and save 3 PM first.
- Repository also blocks queue/final save if 3 PM not saved.

4. Count field validations

Step 3: Non-negative

- Male / Female / Other / Queue cannot be < 0.
- Key: `presiding.count_cannot_be_negative`

Step 4: Max 4 digits

- Any single count field max = 9999.
- Key: `presiding.count_max_four_digits`

Step 5: Cannot exceed login electors

- Male/Female/Other cannot exceed login elector totals.
- Key: `presiding.count_exceeds_electors`

Step 6: Progressive floor (cumulative)

- Later slot cannot be less than earlier saved slot (per gender).
- Example: 9 AM male=40 => 11 AM male >= 40.
- Key: `presiding.count_not_less_than_previous`

5. Last hour + Queue vs Final (STRICT EQUALITY)

Rule: (Last hourly total) + (Queue/katar) MUST EQUAL Final total. Neither less nor greater allowed.

- Urban last hour = 5 PM total (male+female+other).
- Rural last hour = 3 PM total (male+female+other).
- Queue = katar count (single number).
- Final = matdan jankari total (male+female+other).

Step 7: Equality when saving Final

- If Final < (Last + Queue) => REJECT (cannot be less).
- If Final > (Last + Queue) => REJECT (cannot be more).
- If Final == (Last + Queue) => ALLOW.

Step 8: Equality when editing Last/Queue after Final saved

- Same equality rule re-applied.
- Any change making Last+Queue != Final is rejected.

Examples:

- Urban OK: 5PM=100, Katar=5, Final=105
- Urban FAIL: Final=104 or 106
- Rural OK: 3PM=80, Katar=2, Final=82
- Rural FAIL: Final=81 or 83

6. Finish button gate

Step 9: Finish voting and return

- Enabled only when every area-type slot is saved.
- Disabled if any card unsaved, or after already submitted.
- On success: two_hourly_info -> live_poll_info -> poll_end, then back.
- After submit, all turnout cards become read-only.

7. Slot lock after save

- Hourly, queue, final lock after successful save.
- Live poll stays editable until finish boundary.

8. QA recheck checklist

- [] Urban shows 5 PM; Rural does not.
- [] Urban: without 5 PM save, Queue/Final locked + snackbar.
- [] Rural: without 3 PM save, Queue/Final locked + snackbar.
- [] After last hour save, Queue/Final unlock.
- [] Progressive lower-count blocked.
- [] Elector ceiling enforced.
- [] Final != Last+Queue blocked both ways.
- [] Final == Last+Queue saves.

- [] Finish disabled until all saved.
- [] After Finish, fields locked.

9. Code reference

- turnout_count_validator.dart
- presiding_concern_repository_impl.dart
- presiding_turnout_screen.dart
- presiding_turnout_card.dart
- assets/translations/en.json and hi.json

End of document.